$$f(t) = a_0 + \sum_{n=1}^{\infty}(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

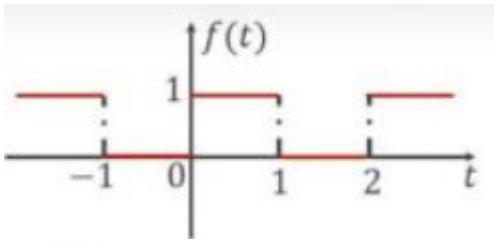
$$A_n\angle\varphi_n = a_n - jb_n \qquad A_n = \sqrt{{a_n}^2 + {b_n}^2}$$

$$a_0 = A_0$$

$$f(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \varphi_n)$$

Phổ tần số của $f(t)$ bao gồm phổ biên độ $A_n = f(n\omega_0)$ và phổ pha $\varphi_n = f(n\omega_0)$.

Ví dụ 1. Tìm các hệ số chuỗi Fourier a_0, a_n, b_n của hàm $f(t)$
Vẽ phổ biên độ và phổ pha



Giải.

$$T = 2(s) \Rightarrow \omega_0 = \frac{2\pi}{T} = \pi\left(\frac{rad}{s}\right)$$

$$f(t) = \begin{cases} 1 \ \text{nếu}\ 0 < t < 1 \\ 0 \ \text{nếu}\ 1 < t < 2 \end{cases}$$

$$a_0 = \frac{1}{2}\int_0^1 1\,dt = 0{,}5$$

$$a_n = \frac{2}{2}\int_0^1 \cos n\pi t\,dt = \frac{1}{n\pi}\sin n\pi t\Big|_0^1 = \frac{1}{n\pi}(\sin n\pi - \sin 0) = 0$$

$$b_n = \frac{2}{2}\int_0^1 \sin n\pi t\, dt = -\frac{1}{n\pi}\cos n\pi t\Big|_0^1$$

$$b_n = \frac{1}{n\pi}(cos\,0 - cos\,n\pi) = \frac{1}{n\pi}(1 - cos\,n\pi)$$

$$b_n = \begin{cases} \dfrac{2}{n\pi} & \text{nếu } n \text{ lẻ} \\ 0 & \text{nếu } n \text{ chẵn} \end{cases}$$

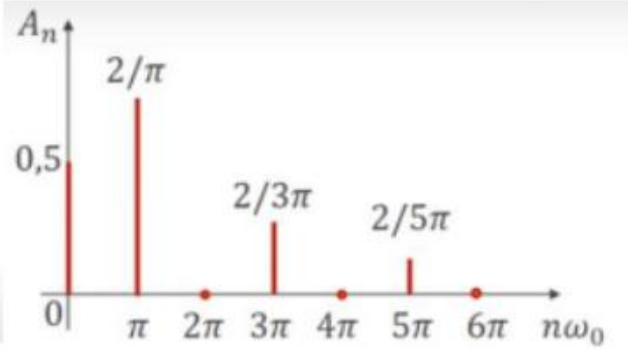
$$f(t) = \frac{1}{2} + \frac{2}{\pi}\sum_{k=1}^{\infty}\frac{1}{n}\sin n\pi t \text{ với } n = 2k - 1$$

phổ biên độ $A_n = f(n\omega_0)$

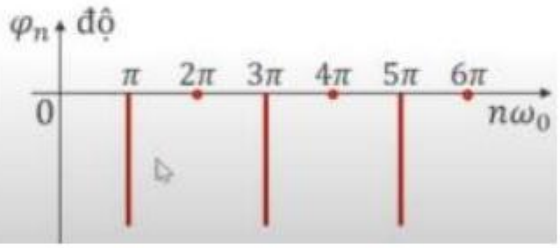
$$a_0 = 0{,}5 = A_0;\ a_n = 0;\ b_n = \begin{cases} \dfrac{2}{n\pi} & \text{nếu } n \text{ lẻ} \\ 0 & \text{nếu } n \text{ chẵn} \end{cases}$$

$$A_n\angle\varphi_n = a_n - jb_n = \begin{cases} -j\dfrac{2}{n\pi} & \text{nếu } n \text{ lẻ} \\ 0 & \text{nếu } n \text{ chẵn} \end{cases}$$

$$A_n = \sqrt{{a_n}^2 + {b_n}^2} = \begin{cases} \dfrac{2}{n\pi} & \text{nếu } n \text{ lẻ} \\ 0 & \text{nếu } n \text{ chẵn} \end{cases}$$

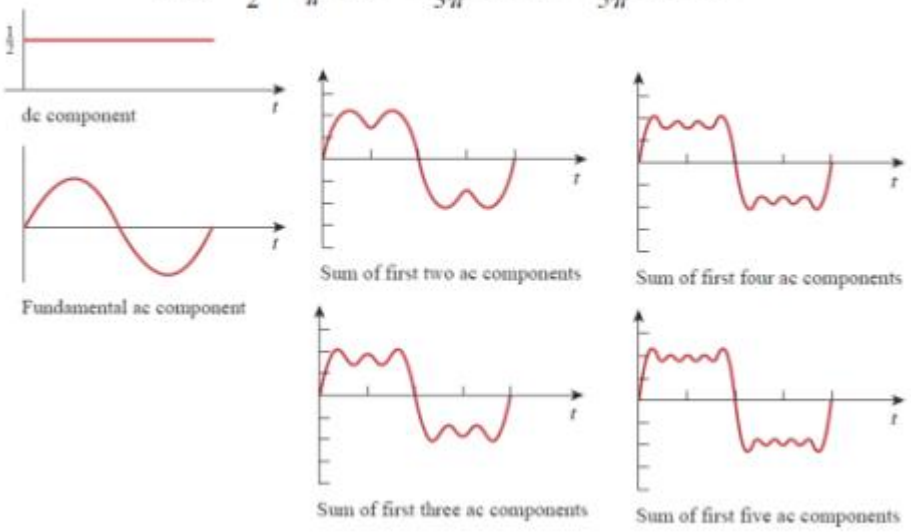


phổ pha $\varphi_n = f(n\omega_0)$



$$\varphi_n = \begin{cases} -90^0 & \text{nếu } n \text{ lẻ} \\ 0 & \text{nếu } n \text{ chẵn} \end{cases}$$

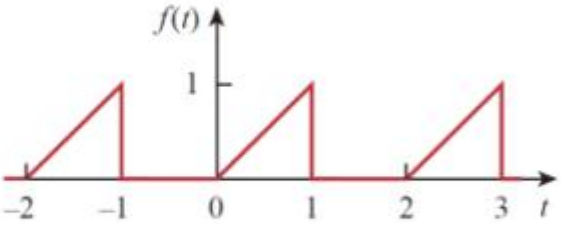
$$f(t) = \frac{1}{2} + \frac{2}{\pi}\sin \pi t + \frac{2}{3\pi}\sin 3\pi t + \frac{2}{5\pi}\sin 5\pi t + \cdots$$



Ý nghĩa xếp chồng : tín hiệu tuần hoàn không sin là tổng của tín hiệu DC và các tín hiệu điều hòa có tần số là bội số của tần số cơ bản.

<u>Ví dụ :</u>

Phân tích dạng sóng sau thành chuỗi Fourier



$$f(t) = \begin{cases} t, & 0 < t < 1 \\ 0, & 1 < t < 2 \end{cases}$$

Giải.

Since $T = 2$, $\omega_0 = 2\pi/T = \pi$. Then

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt = \frac{1}{2}\left[\int_0^1 t\,dt + \int_1^2 0\,dt\right] = \frac{1}{2}\frac{t^2}{2}\Big|_0^1 = \frac{1}{4}$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\,dt$$

$$= \frac{2}{2}\left[\int_0^1 t\cos n\pi t\,dt + \int_1^2 0\cos n\pi t\,dt\right]$$

$$= \left[\frac{1}{n^2\pi^2}\cos n\pi t + \frac{t}{n\pi}\sin n\pi t\right]\Bigg|_0^1$$

$$= \frac{1}{n^2\pi^2}(\cos n\pi - 1) + 0 = \frac{(-1)^n - 1}{n^2\pi^2}$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\,dt$$

$$= \frac{2}{2}\left[\int_0^1 t\sin n\pi t\,dt + \int_1^2 0\sin n\pi t\,dt\right]$$

$$= \left[\frac{1}{n^2\pi^2}\sin n\pi t - \frac{t}{n\pi}\cos n\pi t\right]\Bigg|_0^1$$

$$= 0 - \frac{\cos n\pi}{n\pi} = \frac{(-1)^{n+1}}{n\pi}$$

$$\Rightarrow \quad f(t) = \frac{1}{4} + \sum_{n=1}^{\infty}\left[\frac{[(-1)^n - 1]}{(n\pi)^2}\cos n\pi t + \frac{(-1)^{n+1}}{n\pi}\sin n\pi t\right]$$

Phổ biên độ của tín hiệu

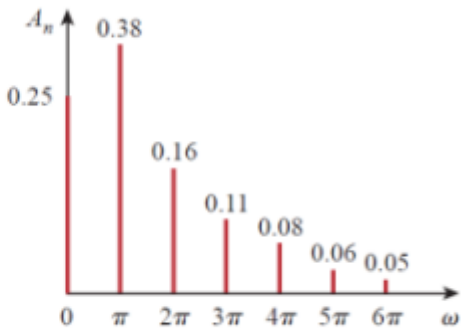
Với n chẵn:

$$A_n = |b_n| = \frac{1}{n\pi}, \qquad n = 2, 4, \ldots$$

Với n lẽ:

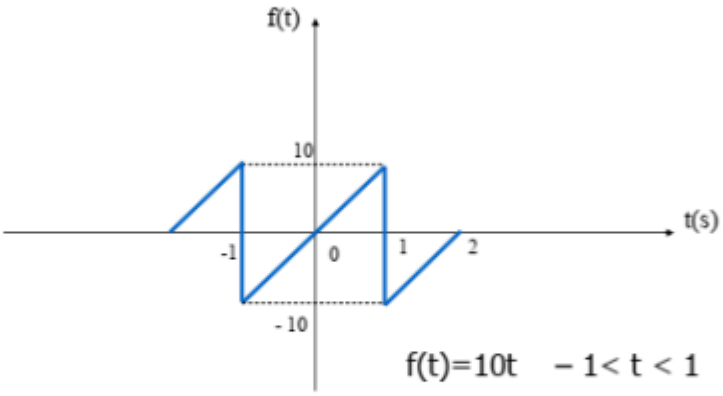
$$A_n = \sqrt{a_n^2 + b_n^2} = \sqrt{\frac{4}{n^4\pi^4} + \frac{1}{n^2\pi^2}}$$

$$= \frac{1}{n^2\pi^2}\sqrt{4 + n^2\pi^2}, \qquad n = 1, 3, \ldots$$



Ví dụ :

Phân tích dạng sóng sau thành chuỗi Fourier



f(t)=10t $-1 < t < 1$

Giải • T = 2 ; $\omega_0 = 2\pi/T = \pi$. Nhìn vào dạng sóng ta thấy hàm đối xứng lẻ

Vậy $a_0 = 0$ và $a_n = 0$

•Xác định b_n:

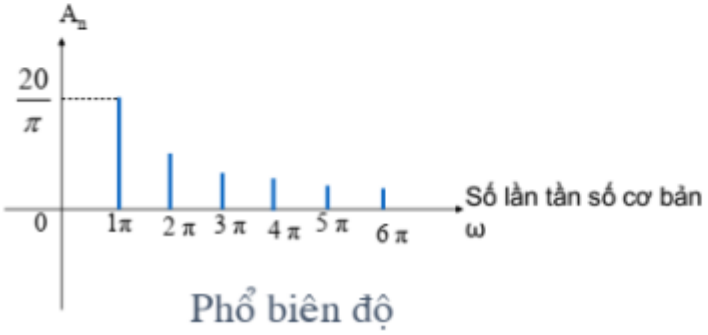
$$b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin(n\omega_0 t)dt = 2\int_0^1 10t\sin(\pi nt)dt = 20\int_0^1 t\sin(\pi nt)dt$$

$$= 20\left[\frac{1}{(\pi n)^2}\sin(n\pi t) - \frac{1}{n\pi}t\cos(n\pi t)\right|_0^1 = -\frac{20\cos(n\pi)}{n\pi}$$

• n lẻ: $b_n = \frac{20}{n\pi}$ $b_1 = \frac{20}{\pi}$ $b_3 = \frac{20}{3\pi}$ $b_5 = \frac{20}{5\pi}$

• n chẵn: $b_n = -\frac{20}{n\pi}$ $b_2 = -\frac{10}{\pi}$ $b_4 = -\frac{5}{\pi}$

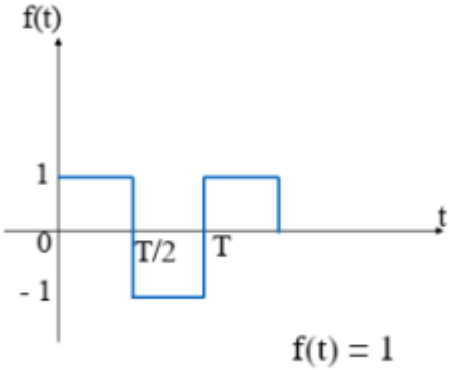
$$f(t) = \frac{20}{\pi}(\sin\pi t - \frac{1}{2}\sin 2\pi t + \frac{1}{3}\sin 3\pi t - ...)$$



Phổ biên độ

Ví dụ :

Phân tích dạng sóng sau thành chuỗi Fourier, có biên độ là 1 và chu kỳ là 2π.



$f(t) = 1 \qquad 0 < t < T/2$

$f(t) = -1 \qquad T/2 < t < T$

Giải

$$T = 2\pi \rightarrow \omega_0 = \frac{2\pi}{T} = 1(\text{rad/s})$$

Chuỗi Fourier có dạng :

$$f(t) = a_0 + \sum_{n=1}^{\infty}(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Xác định a_0:

$$a_0 = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)dt = \frac{1}{T}\int_0^T f(t)dt = \frac{1}{T}\left[\int_0^{T/2} 1dt + \int_{T/2}^{T}(-1)dt\right]$$

$$= \frac{1}{T}\left[t\Big|_0^{T/2} - t\Big|_{T/2}^{T}\right] = \frac{1}{T}\left[\frac{T}{2} - \frac{T}{2}\right] = 0$$

•Xác định a_n, b_n:

$$a_n = \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\cos n\omega_0 t dt = \frac{2}{T}\left[\int_0^{T/2}\cos n\omega_0 t dt - \int_{T/2}^{T}\cos n\omega_0 t dt\right]$$

$$= \frac{2}{T}\left[\frac{1}{n\omega_0}\sin n\omega_0 t\Big|_0^{T/2} - \frac{1}{n\omega_0}\sin n\omega_0 t\Big|_{T/2}^{T}\right] = 0$$

$$b_n = \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\sin n\omega_0 t dt = \frac{2}{T}\left[\int_0^{T/2}\sin n\omega_0 t dt - \int_{T/2}^{T}\sin n\omega_0 t dt\right]$$

$$= \frac{2}{T}\left[-\frac{1}{n\omega_0}\cos n\omega_0 t\Big|_0^{T/2} + \frac{1}{n\omega_0}\cos n\omega_0 t\Big|_{T/2}^{T}\right]$$

$$= \frac{2}{n2\pi}(-\cos n\pi + 1 + 1 - \cos n\pi) = \frac{2}{n\pi}(1 - \cos n\pi)$$

$\cos n\pi = -1$ Khi n lẻ (2n-1)

$\cos n\pi = +1$ Khi n chẵn (2n)

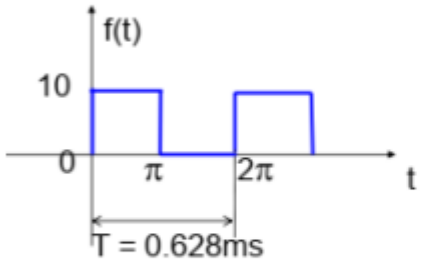
Khi n lẻ: $b_n = \frac{2}{(2n-1)\pi} * 2 = \frac{4}{(2n-1)\pi}$

Khi n chẵn: $b_n = 0$

Vậy: $\boxed{f(t) = \frac{4}{\pi}(\sin t + \frac{1}{3}\sin 3t + \frac{1}{5}\sin 5t + ...)}$

Ví dụ :

Phân tích dạng sóng sau thành chuỗi Fourier



f(t) = 10 $\quad 0 < t < \pi$

f(t) = 0 $\quad \pi < t < 2\pi$

Giải

$T = 2\pi$; $\omega_0 = 2\pi/T = 1$

• Xác định a_0:

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(t)dt$$

$$a_0 = \frac{1}{2\pi}\left[\int_0^{\pi} 10dt + \int_{\pi}^{2\pi} 0dt\right] = 5$$

• Xác định a_n:

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(t).\cos nt\, dt$$

$$a_n = \frac{1}{\pi}\left[\int_0^{\pi} 10.\cos nt\, dt + \int_{\pi}^{2\pi} 0.\cos nt\, dt\right] = \frac{10}{\pi n}\sin nt\Big|_0^{\pi} = \frac{10}{\pi n}\sin n\pi$$

Ta thấy $a_n = 0$ với n = 1, 2… $(a_1, a_2, \ldots, a_n = 0)$

• Xác định b_n:

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(t).\sin nt\, dt$$

$$b_n = \frac{1}{\pi}\left[\int_0^{\pi} 10.\sin nt\, dt + \int_{\pi}^{2\pi} 0.\sin nt\, dt\right] = \frac{10}{\pi n}(-\cos nt)\Big|_0^{\pi} = \frac{10}{\pi n}(1 - \cos n\pi)$$

❖Khi n lẻ: $b_n = \frac{20}{n\pi}$ $b_1 = \frac{20}{\pi}$ $b_3 = \frac{20}{3\pi}$ $b_5 = \frac{20}{5\pi}$.....

❖Khi n chẵn: $b_n = 0$

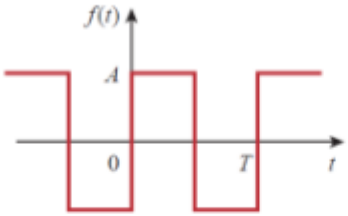
$$f(t) = 5 + \frac{20}{\pi}(\sin t + \frac{1}{3}\sin 3t + \frac{1}{5}\sin 5t + \ldots)$$

Khi T = 0,628ms ⇒ f = 1592,36Hz ⇒ ω = 2πf = 10000 rad/s

$$f(t) = 5 + \frac{20}{\pi}(\sin 10000t + \frac{1}{3}\sin 30000t + \frac{1}{5}\sin 50000t + \ldots)$$

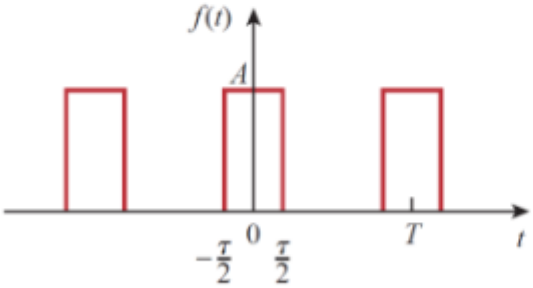
Chuỗi Fourier của một vài dạng sóng cơ bản

1. Square wave



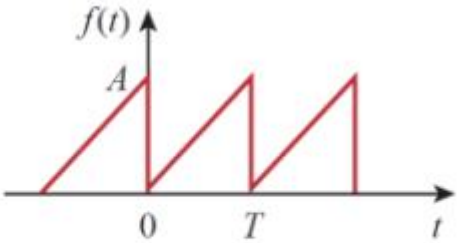
$$f(t) = \frac{4A}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin(2n-1)\omega_0 t$$

2. Rectangular pulse train



$$f(t) = \frac{A\tau}{T} + \frac{2A}{T} \sum_{n=1}^{\infty} \frac{1}{n} \sin\frac{n\pi\tau}{T} \cos n\omega_0 t$$

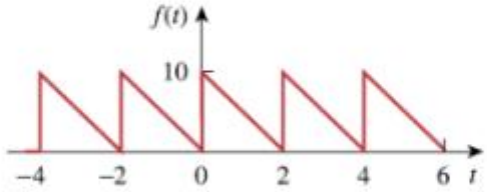
3. Sawtooth wave



$$f(t) = \frac{A}{2} - \frac{4A}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)\omega_0 t$$

BÀI TẬP 1

Phân tích dạng sóng sau thành chuỗi Fourier



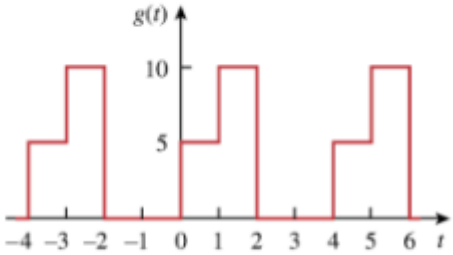
$f(t) = 10 - 5t,\ 0 < t < 2,\ T = 2,\ \omega_o = 2\pi/T = \pi$

ĐÁP ÁN

$$f(t) = 5 + \frac{10}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin(n\pi t)$$

BÀI TẬP 2

Phân tích dạng sóng sau thành chuỗi Fourier



$T = 4,\ \omega_o = 2\pi/T = \pi/2$

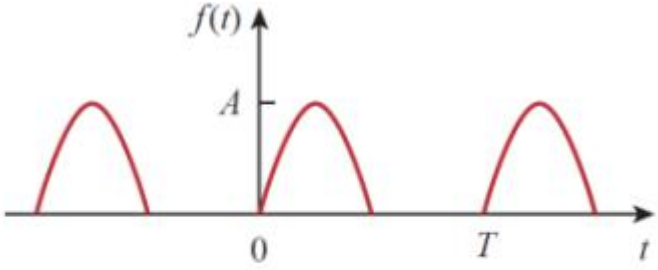
$$g(t) = \begin{cases} 5, & 0 < t < 1 \\ 10, & 1 < t < 2 \\ 0, & 2 < t < 4 \end{cases}$$

ĐÁP ÁN

$$f(t) = 3,75 + \sum_{k=1}^{+\infty}\frac{5}{n\pi}(-1)^{\frac{n+1}{2}}\cos n\frac{\pi}{2}t + \sum_{k=1}^{+\infty}\frac{25}{n\pi}\sin n\frac{\pi}{2}t$$

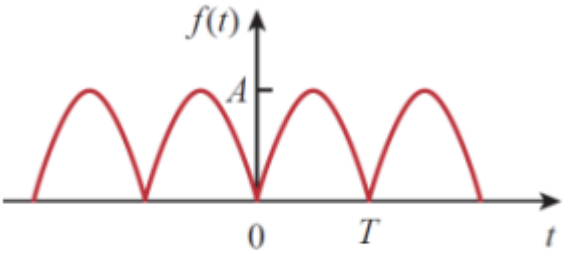
$$+\sum_{i=1}^{+\infty}\frac{5}{n\pi}\left[1+(-1)^{\frac{n}{2}}\right]\sin n\frac{\pi}{2}t \qquad n = 2k-1; n = 2i$$

BÀI TẬP 3



$$f(t) = \frac{A}{\pi} + \frac{A}{2}\sin\omega_0 t - \frac{2A}{\pi}\sum_{n=1}^{\infty}\frac{1}{4n^2-1}\cos 2n\omega_0 t$$

BÀI TẬP 4



$$f(t) = \frac{2A}{\pi} - \frac{4A}{\pi}\sum_{n=1}^{\infty}\frac{1}{4n^2-1}\cos n\omega_0 t$$